

Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka

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Our Team



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Problem Statement & Stakeholders



Research Problem

Sri Lankan middle school STEM education is mostly textbook-based and theory-driven, with limited opportunities for practical experimentation.



Stakeholders

- **Primary Users** - Grade 7–11 students
- **Secondary Users** - Teachers
- **Decision Makers** - Schools and Institutes



Observed Issue

Students struggle to visualize abstract STEM concepts and connect theory with real-world applications.



Why It Matters

- Low student engagement
- Poor conceptual understanding
- Limited interest in STEM careers

Research Gap

Feature	Existing Systems	Proposed System
Sri Lankan syllabus alignment	✗	✓
Hands-on VR task simulation	✗	✓
Gamified learning progression	⚠	✓
Practical skill assessment	✗	✓
Sinhala/English support	✗	✓
Offline school usage	✗	✓

Most VR education tools are not aligned with the Sri Lankan curriculum

Many systems provide passive experiences (videos or demonstrations) instead of interactive learning

Existing platforms lack cultural and historical context

Limited gamification and immersive interaction

Objectives

Main Objective



Develop a Gamified Virtual Reality Educational Platform that delivers STEM learning activities aligned with the Sri Lankan middle school curriculum.

Key Objectives



- Provide curriculum-aligned educational tasks in a culturally relevant environment
- Improve student engagement using immersive VR learning
- Visualize abstract STEM concepts through interactive simulations
- Integrate gamification elements such as challenges and rewards



Solution Overview

Solution Summary

A Gamified VR Educational Platform that enables Sri Lankan middle school students to learn STEM concepts through immersive simulations in a historical virtual environment.

Key Features

- Interactive VR learning environment
- Gamified tasks and challenges
- Curriculum-aligned STEM activities
- Real-time feedback and progress tracking
- Historical Sri Lankan village simulation





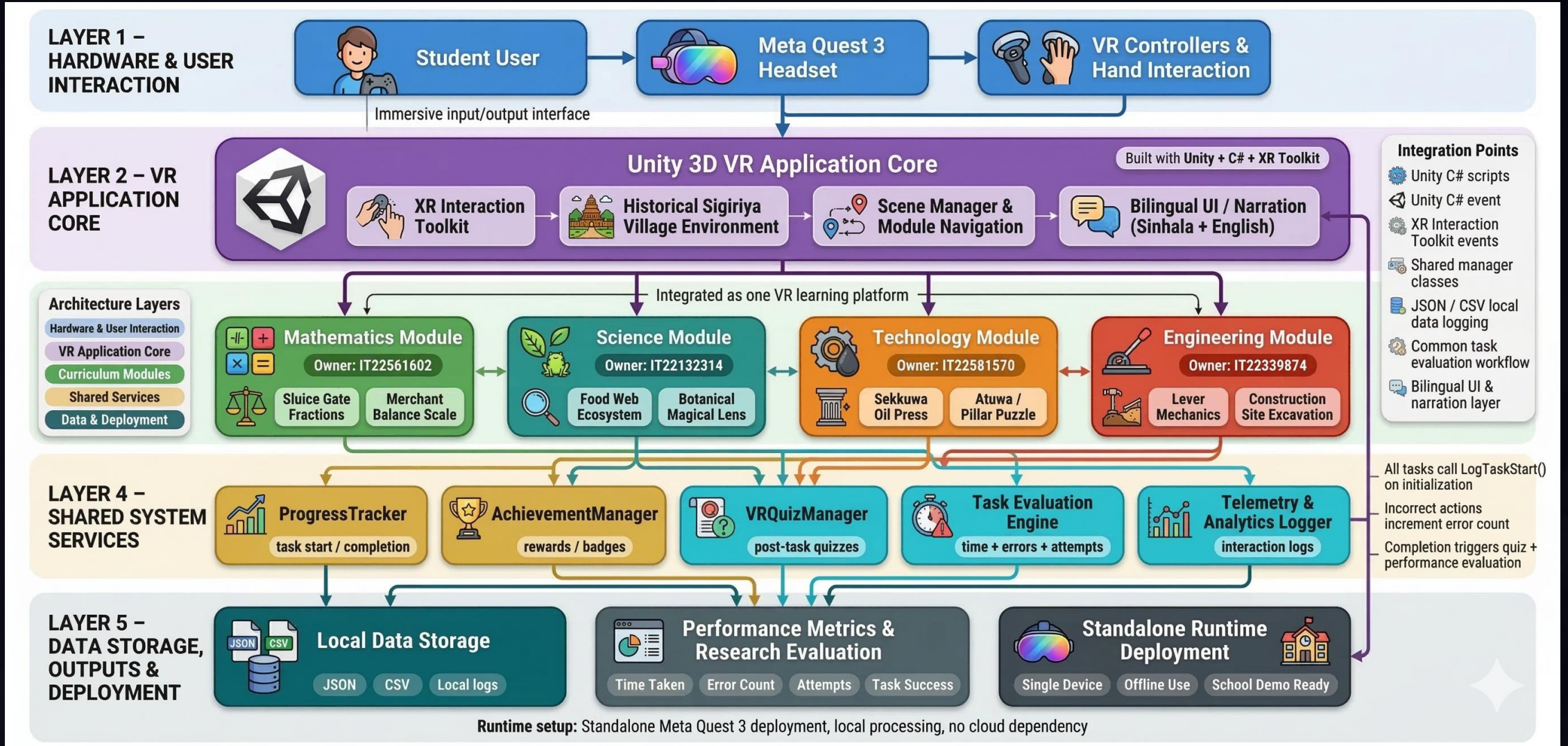
Non-Functional Focus

- High VR performance
- User-friendly interaction using VR controllers
- Bilingual instruction support
- Reliable real-time interaction tracking

Current Completion Status

- ✓ VR environment implemented
- ✓ Core learning modules integrated
- ✓ Interaction mechanics developed
- ✓ Evaluation tracking system implemented

System Architecture & Integration



Commercialization / Impact

Target Users / Market

- Middle school students (Grades 7–11)
- STEM subject teachers
- Sri Lankan schools



Value Proposition



Immersive hands-on STEM learning



Gamified learning that increases engagement



Curriculum-aligned VR simulations



Safe virtual environment for practical experiments

Cost & Feasibility

Item	Estimated Cost (LKR)	Notes	Recovery / Pricing Strategy
Development per module (Math, Science, English, History)	5,000 – 10,000	Covers 3D assets, narration, testing, deployment	Cost recovered through annual school subscriptions
Total development (4 modules)	20,000 – 40,000	Full platform build excluding hardware	Breakeven within 2–3 years depending on adoption scale
Hardware (Meta Quest 3 headset)	~180,000	Primary classroom device, one per school initially	Shared across subjects, one-time investment
Other expenses (travel, teacher training, contingency)	~20,000 – 30,000	Pilot deployments and support	Covered through subscription margin or NGO funding
Suggested pricing	10,000 – 15,000 per school / year	Affordable for schools, scalable with adoption	Subscription and update fees ensure recurring revenue

Adoption Plan



Pilot testing in selected schools



Teacher training programs



Integration with STEM curriculum activities



Gradual rollout through education technology initiatives

Demo Plan

- 1 System Introduction
- 2 Mathematics Module Learning Tasks
- 3 Science Module Learning Tasks
- 4 Technology Module Learning Tasks
- 5 Engineering Module Learning Tasks



Demo Responsibility

Member	Demo Section
IT22561602 – Dayananda W.C.R	System Introduction + Mathematics Module
IT22132314 – Wijenarayana C.H	Science Module
IT22581570 – Jayalath K.M.K.D	Technology Module
IT22339874 – Ubeywarna R.C	Engineering Module



Thank You!

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